Activity #8: Bahrain Open Data Table Project

For this activity, you will use the Bahrain Open Data Portal at **data.gov.bh** to choose **any available dataset**, retrieve its data using the free API, and present the results in a clean HTML table using **JavaScript fetch**. The Bahrain portal provides public datasets and an API console, and its API returns JSON data over HTTP GET, so students do not need an API key for this task.

Activity goal

Create a webpage that:

- connects to **https://www.data.gov.bh/**
- uses **one dataset of your choice**
- retrieves records using **fetch()**
- displays the data inside an HTML table
- handles the data cleanly and professionally

Important rule

You do **not** need an API key for this activity because the Bahrain Open Data site provides open public access to its datasets and API.

Student task

Build your own HTML page from scratch and use JavaScript to fetch data from **one dataset** on the Bahrain Open Data Portal.

You may choose any dataset you like from the portal's Explore section, which includes many categories such as weather, population, transport, social data, and economic indicators.

Your page must show the dataset records in a properly formatted HTML table.

Guided instructions

Step 1: Open the Bahrain Open Data website

Go to:

https://www.data.gov.bh/

Browse the available datasets and choose one that looks suitable for displaying in a table. The portal includes an Explore section and dataset pages with table, export, and API options.

Hints

- Choose a dataset with clear columns and multiple rows.
- Avoid choosing a dataset that is too complex for your first attempt.
- Try to find one where the column names are easy to understand.

Step 2: Find the dataset API

After choosing a dataset, open its page and look for the **API** option. The Bahrain Open Data Portal provides an API console for datasets through its Opendatasoft Explore API.

Hints

- Look for buttons or tabs such as **API**, **Export**, or **Table**.
- You need the API endpoint that returns **JSON**.
- Make sure you are using a records endpoint, not a webpage URL.

Think carefully

- Are you copying the dataset page link, or the actual API link?
- Does the URL return JSON data when opened?

Step 3: Create the HTML page

You must build an HTML page that includes:

- a page title
- a heading showing the dataset name

- a button or automatic load feature
- a table with a header row
- a table body that will be filled using JavaScript

Hints

- Use <table>, <thead>, and <tbody>
- Keep the table readable
- Add borders or Bootstrap styling if you want

Step 4: Use fetch() in JavaScript

Use JavaScript fetch() to request the dataset data from the API.

Hints

- Use async and await
- Convert the response to JSON
- Inspect the JSON structure carefully before building the table
- Find the array that contains the records

Think carefully

- Where are the actual records stored in the response?
- Is the data directly in the root object, or inside another property?

Step 5: Build the table dynamically

Once you receive the JSON data, loop through the records and create rows for the table.

Hints

- Use a loop to go through each record
- Choose a few useful columns from the dataset
- Insert the values into table cells
- Update the <tbody> using JavaScript

Think carefully

- Which fields are the most useful to show?
- Are some field names too long or unclear for table headers?

Step 6: Handle missing values

Some datasets may have empty or missing fields.

Hints

- Do not assume every value exists
- Show N/A or Not Available when data is missing
- Make sure your page does not break if one field is empty

Step 7: Make the table look professional

Your table should be neat and easy to read.

Hints

- Add a heading with the dataset title
- Use clear column headings
- Keep data aligned properly
- Make the table readable on screen

Optional improvements

- Add a loading message
- Add a “Load Data” button
- Add a record counter
- Show an error message if the fetch fails

What students must submit

Each student should submit:

- one HTML file
- embedded or linked JavaScript
- a working table that displays data from **one selected dataset** on **data.gov.bh**
- the dataset name written clearly on the page

Rules for this activity

- Use **https://www.data.gov.bh/** only
- Choose **any available dataset**
- Use **fetch()** in JavaScript
- Show the results in a table
- Do **not** copy code from others
- Do **not** paste the raw JSON only; it must be displayed in a proper HTML table

Activity #8 - Rubric

Criteria	Excellent (Full Marks)	Good	Basic	Limited / Missing	Marks
1. Dataset Selection & API Usage (4 marks)	Correct dataset chosen, proper API endpoint used, data successfully retrieved	Minor issues but data loads correctly	API works partially or incorrect endpoint used	No working API connection	/4
2. Fetch & Data Retrieval (4 marks)	Proper use of fetch() with correct JSON handling and structure understanding	Minor issues but data is retrieved	Basic fetch implemented but errors in handling data	Fetch not implemented or not working	/4
3. Table Structure & Data Display (4 marks)	Data clearly displayed in a well-structured table with correct columns	Table works with minor formatting issues	Table displays some data but incomplete or unclear	Table missing or not functional	/4
4. Handling Missing/Null Data (3 marks)	Proper handling of missing values with clean fallbacks (e.g., N/A)	Some handling but inconsistent	Minimal handling of missing data	No handling of missing data	/3
5. User Interface & Presentation (3 marks)	Clean, readable, well-formatted table with clear headings	Mostly clear with minor design issues	Basic layout with limited readability	Poor or unclear presentation	/3
6. Code Quality & Organization (2 marks)	Clean, well-organized, readable code with good structure	Mostly clear with minor issues	Some structure but messy or inconsistent	Poor structure or hard to read	/2

Total: /20